

# OpenWebSearch.eu — Project Overview

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Overview of goals, results, impact, and outlook

# Overall Goal

Build and pilot the core of a **European Open Web Index (OWI)** and a federated **Open Web Search and Analysis Infrastructure** at TRL 5.

## Project proposition

Reduce the entry costs for European researchers, innovators, and organisations that want to use the Web for search, analytics, and AI.

## Operational interpretation

Build not only a dataset, but a reusable stack for **crawling, enrichment, indexing, distribution, and retrieval**.



# Project Objectives I

## 1. Open technology stack

- open pipelines for crawling, preprocessing, indexing, and access
- reusable components instead of one closed vertical service
- basis for search, analytics, and AI-oriented retrieval

## 2. Resource provision

- federated deployment across European HPC and data-centre infrastructure
- shared storage, orchestration, and publishing capabilities
- technical basis for future scaling via EuroHPC and AI Factory contexts



These two objectives define the infrastructural core of the project: without them, there is no operational OWI.

# Project Objectives II

## 3. Added-value services

- dashboards, access tools, APIs, and prototype search services
- transparency and webmaster-facing services
- lower barriers for external use of the OWI

These objectives matter because an open index is only useful if it is accessible, governable, and used.

## 4. Ecosystem building

- applications, community programme, governance, and exploitation work
- legal and ethical framing for responsible use
- uptake beyond the original consortium

# Core Result: OWI, Infrastructure, Community

## Open Web Index

- **39.6B** fetched pages
- **12.6B** unique pages
- **10B+** indexed pages
- **500+** daily snapshots

Detailed results: **WP1-WP3**

## Federated Infrastructure

- **1.25 PiB** crawled raw data
- workflows across **3 EuroHPC JU centres + 3 further data centres**
- LEXIS, iRODS, S3, HEAppE as operational backbone

Detailed results: **WP5**, plus **WP1-WP3**

## Community & Uptake

- **180+** community members
- **16** third-party projects funded
- governance and continuation work established

Detailed results: **WP6-WP8**



# The OWI as the Technical Core

## What was delivered

- **39.6B** fetched pages
- **12.6B** unique pages
- **1.25 PiB** crawled raw data
- **10B+** indexed pages
- **75M domains** and **200M hosts**
- **500+ daily snapshots** in ClIFF + Parquet
- **58 public OWIE datasets** with **119M embeddings**

## Why this is the core result

- Europe now has a working, open, federated web-index backbone instead of only a concept
- the pipeline is operational end to end: **crawl -> preprocess -> enrich -> index -> distribute -> search**
- the index is not only downloadable; it is also **queryable and application-ready**
- the project demonstrated that open web search infrastructure can run across European HPC and cloud environments

Most important technical message: an **Open Web Index at meaningful scale is feasible.**



Detailed results in **WP1** crawling, **WP2** enrichment, **WP3** indexing and access, **WP5** infrastructure.

# Main Results by Work Package

## WP1-WP2

- federated crawling
- aggregated crawl transparency
- webmaster console
- preprocessing, content analysis, embeddings

## WP3-WP5

- ClIFF + Parquet index production
- remote index, OWIlix, OURRS
- federated storage and workflow execution

## WP4

- Mosaic, MosaicRAG, EO search, argument search, recommender
- user-centred evaluation of applications

## WP6-WP8

- legal, ethical, and societal groundwork
- dissemination, exploitation, and community building
- coordination, FSTP, governance, and continuation planning

# Work Package Overview

| WP  | Outcome at review level                                                                             |
|-----|-----------------------------------------------------------------------------------------------------|
| WP1 | Federated crawling, OWSI/OCSI aggregation, Open Webmaster Console, structured harvesting            |
| WP2 | Scalable preprocessing, content analysis, nutrition labels, geoparsing, embeddings                  |
| WP3 | Index production, remote index, OWIlix, OURRS, evaluation collections                               |
| WP4 | Search applications and user evaluation: Mosaic, MosaicRAG, EO search, argument search, recommender |
| WP5 | Federated data infrastructure across European compute and storage sites                             |
| WP6 | Legal, ethical, societal, governance, sustainability groundwork                                     |
| WP7 | Dissemination, exploitation, communication, community growth                                        |
| WP8 | Coordination, FSTP execution, policy engagement, post-project continuation setup                    |

The technical WPs produced the backbone; the non-technical WPs made it usable, discussable, and continuable.



# Impact

## Strategic impact

- reduces dependency on proprietary, non-European web indices
- creates a shared resource for search, analytics, and AI experimentation
- establishes an operational model relevant for **AI Factories**, **EOSC**, and data spaces

## Quantifiable signals

- host coverage target of **500M** exceeded with **516M hosts**
- **180+** community members
- **93 papers** published in RP2
- estimated **~EUR 4.5B** net economic and societal benefits over a decade

## Policy and ecosystem impact

- open web search entered discussions with the EC, EuroHPC, and the European Parliament
- three #ossym symposia and three WOWS workshops helped establish the field
- public launch of the OWI in June 2025
- governance work converged on a **MoU-based transition model**

## Realistic assessment

- strong proof of feasibility
- important services and assets now exist
- not yet a finished product or 24/7 service
- main remaining gaps are funding, operations, and legal-regulatory conditions

# Sustainability: MoU-Based Continuation

## Why an MoU

- the project created infrastructure, data products, and community assets that should not end with the funded pilot
- a permanent legal entity was assessed, but not yet realistic as an immediate transition step
- the **Memorandum of Understanding** is the most practical bridge between project end and longer-term institutionalisation

## What the MoU enables

- coordinated continuation of core OWI operations on a best-effort basis
- shared governance and clear responsibilities among continuing partners
- continuity for community, infrastructure, and service access while follow-up structures mature

## What it does not solve on its own

- no guaranteed base funding
- no automatic move to full production-grade 24/7 operations
- no resolution of the wider legal and regulatory barriers for commercial exploitation

## Why it still matters

- it keeps the project legacy actionable rather than symbolic
- it creates a credible handover model for EuroHPC / AI Factory related uptake
- it gives time to move from pilot coordination to durable European governance

# Follow-Up on Previous Review Recommendations

## 1. Ethical / legal implications of making the index available

- ethical self-assessment, takedown handling, transparency mechanisms, webmaster controls
- legal studies resulted in an **Open Web Index License** for scientific use
- policy recommendations were produced and communicated to EU policy processes

## 2. Sustainability and exploitation after project end

- Open Search Foundation coordination plus a **MoU-based best-effort continuation**
- governance options assessed; transitional structure defined
- follow-up projects secured, including **SOURCE** starting in **April 2026**
- AI Factory / AI Factory Antenna discussions now provide a realistic scaling path

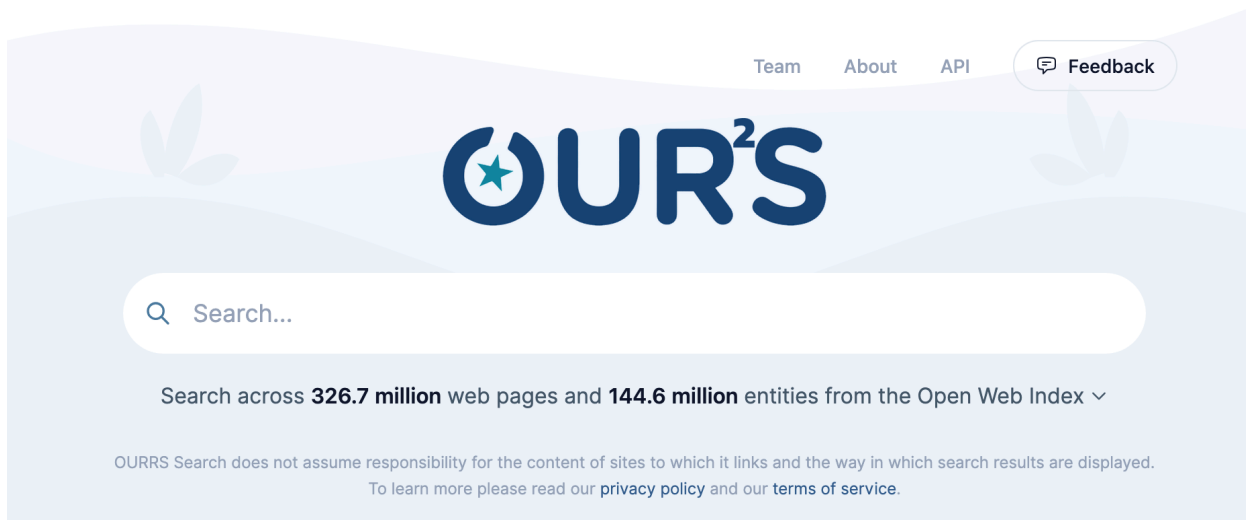
## 3. Adapting to LLM / GenAI developments

- Resilipipe GPU added large-scale embedding generation
- **German Commons** and **WebFAQ** created for European LLM work
- model and dataset hosting capabilities added to the platform

≡ **MOSAIC-RAG** and **OURRS** demonstrate practical RAG-oriented retrieval on top of OWI

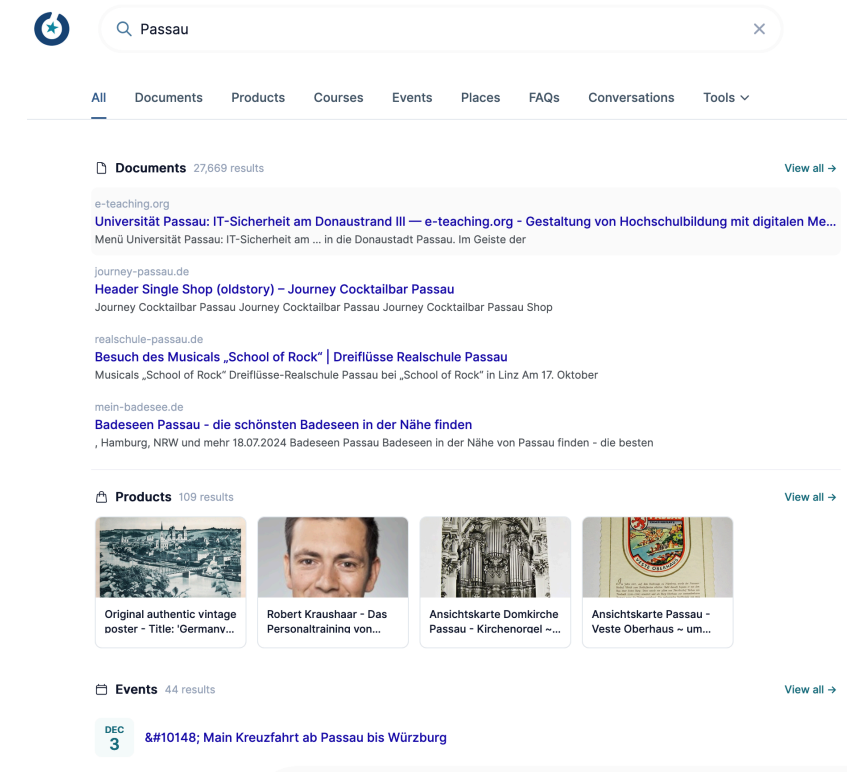
Detailed results in **WP2**, **WP3**, **WP6**, **WP7**, and **WP8**.

# Demo: OURRS as the User-Facing Proof Point



## Why this demo matters

- shows OWI is not only a dataset archive
- provides a hosted search UI and API over OWI content
- lowers adoption barriers for application developers and RAG systems



## Demo flow

1. issue a query against live OWI-backed search
2. inspect retrieval quality and result transparency
3. connect the demo back to the pipeline and access story

Claim to demonstrate: **Open infrastructure can support usable retrieval services, not just offline research**

# Final Takeaway

## Delivered

- a federated European Open Web Index at meaningful scale
- the end-to-end open pipeline to build and refresh it
- access modes for datasets, remote querying, and live search
- applications, community, governance, and policy traction

## What matters most after M42

- the project de-risked the core technical question
- the next challenge is no longer "can this be built?"
- it is now "how does Europe sustain and scale it?"

OpenWebSearch.eu turned European open web search from an idea into operating infrastructure.